

So you're thinking about a trade

The short version — written for you, not about you.

Everything that matters, none of the bits written for your parents.

We scored twenty-eight careers on how exposed they are to AI.

A service electrician: 2.5 out of 10 — one of the lowest scores we found anywhere.

An entry-level software developer: 8.1.

And the electrician has no student debt and started earning four years earlier.

The 60-second version

- The trades are among the most protected work we measured. That part isn't close.
 - The way in is getting easier while most degree routes get harder.
 - No tuition. You're paid from day one.
 - The protection isn't "manual work." It's unpredictable work — and that distinction decides everything.
 - Electrical is the standout, and the reason is AI itself.
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Why it's protected

Four reasons, and none of them depend on how clever AI gets.

You have to physically be there, in a different building every time, with unlabelled wiring and work someone did badly in 1974.

The law requires a licence for electrical and gas work. Inspectors sign things off. Nobody approves their own unlicensed work.

Someone's letting you into their home. That's trust, and it's the foundation of every trade business.

Every job is genuinely new. The environment is different every single time, which is exactly the condition robots handle worst.

The thing most people get wrong

You'll hear "manual work is safe from AI." That's not quite right, and the difference matters for what you choose.

What actually protects you isn't that the work is physical. It's that the environment is unpredictable.

Robots are good in controlled settings — known dimensions, repeated tasks, no surprises. A warehouse. An assembly line. A modular construction factory.

They're bad in uncontrolled ones. A crawlspace. A loft. A sixty-year-old house where nothing matches the drawings.

Look at the difference in our own numbers:

SETTING	SCORE
Electrician — service & repair	2.5
Plumbing / HVAC — service	2.6
Welding — site work	3.2
Construction site trades	3.4
Production line / prefab	5.0

Same skills. Twice the exposure, purely because the factory is predictable.

So the question to ask about any hands-on job is: would two days of this ever look the same?
If no, you're protected. If yes, less than you think.

The electrical thing

This is the strangest finding in the whole index.

AI data centres are being built at record pace. They need enormous amounts of electrical infrastructure.

Microsoft's president has said publicly that the electrician shortage is the number-one thing slowing data centre expansion. Not chips. Not land. Not permits. Electricians.

Industry estimates put the need at 300,000-plus additional electricians just for AI-related demand. Electricians working on data centre projects are reporting earnings well into six figures.

So the same technology squeezing white-collar entry jobs is creating a shortage of electricians. That's not irony — it's the clearest possible illustration of where the protection actually sits.

The money, honestly

Nobody runs this comparison properly, so here it is.

BLS medians: electricians \$62,350, plumbers \$62,970. Top earners in both exceed \$106,000.

You earn while you train — typically starting around \$15–20 an hour, stepping up every year. Four to five years to journeyman.

Union total compensation runs about 60% higher than non-union once health and pension are counted.

And construction wages grew 4.2% year-over-year into mid-2025, faster than the national average.

Now the comparison. A graduate finishes at 22 with debt, into a market where 41.5% of recent graduates are working jobs that don't need a degree. You finish at 22 or 23 with no debt, four years of earnings behind you, a licence, and employers competing for you.

The top end isn't the median, though. Getting to \$100,000-plus usually means running your own business — which is a different skill from the trade, and worth knowing about now.

What we won't soften

Your body is the real cost. Knees, backs, shoulders. Heat, cold, confined spaces, awkward positions. It's safer than it was — the injury rate for specialty trades is 2.3 per 100 workers and falling — but it's still harder on a body than any desk job, and thirty years of it adds up.

The ceiling is lower without ownership. Solid money, good money with specialisation, great money if you run something. That last step is a business decision, not a trade one.

It's cyclical. Construction rises and falls with the economy in a way that nursing and teaching don't.

And some people will be condescending about it. That says more about them than about the numbers, but it's real and worth being ready for.

Does this actually sound like you?

It suits people who:

- **Want the problem solved by the end of the day** — the single most-cited thing people love
- Would rather be **moving than sitting**
- Like **diagnosis** — the puzzle of finding what's actually wrong
- Can **deal with people**, since most trades involve customers
- Might want to **run something** eventually

It's harder if you genuinely dislike physical discomfort — being outside in February isn't a phase you grow out of — or if you need predictable indoor conditions, or want a clear ladder of titles to climb.

What actually matters when picking an apprenticeship

- **Completion rate.** The number that matters most and the one least advertised. Ask directly.
- **Breadth of work** — domestic only, or commercial and industrial too? Breadth sets your ceiling.
- **Is there a job at the end**, and what proportion stay on?
- **Does it lead cleanly to the licence**, or are there gaps you'll have to fill yourself?
- **Who's actually supervising you.** In an apprenticeship, the person teaching you is the education.

And choose the trade deliberately. Electrical leads on pay, growth and protection, and it's sitting directly in front of the demand wave.

Two things worth doing

Spend a full day with someone working. Not a college open day — a real working day, including the boring and unpleasant parts. It'll tell you more than anything else you could do.

Then find someone ten years in and ask how their body feels, and whether they'd do it again. Ask someone who'll be straight with you.

One last thing

If someone has treated this as a fallback — a thing you do if the grades don't work out — it's worth knowing that the numbers don't support that view.

Better protection than most degrees. No debt. Earlier earnings. Stronger demand. A realistic route to running your own business.

It isn't a consolation prize. It's also not for everyone, and the physical reality is the part to be honest with yourself about. But the decision deserves to be made on what it actually is, not on what people assume it is.

If you want to go further — three things worth twenty minutes

- [**Electricians — Occupational Outlook Handbook**](#) — the government's own data. Growth, pay, entry requirements. Look up plumbing and HVAC while you're there.
 - [**Electrician career guide 2026**](#) — the data centre story, and why Microsoft says electricians are the bottleneck.
 - [**Skilled trades statistics for 2026**](#) — wages, injury rates and union pay differences in one place.
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There's a longer version behind this — full scores by trade and setting, how they were worked out, the honest downsides, what to ask an apprenticeship provider, and where we might be wrong. Your parents have it.